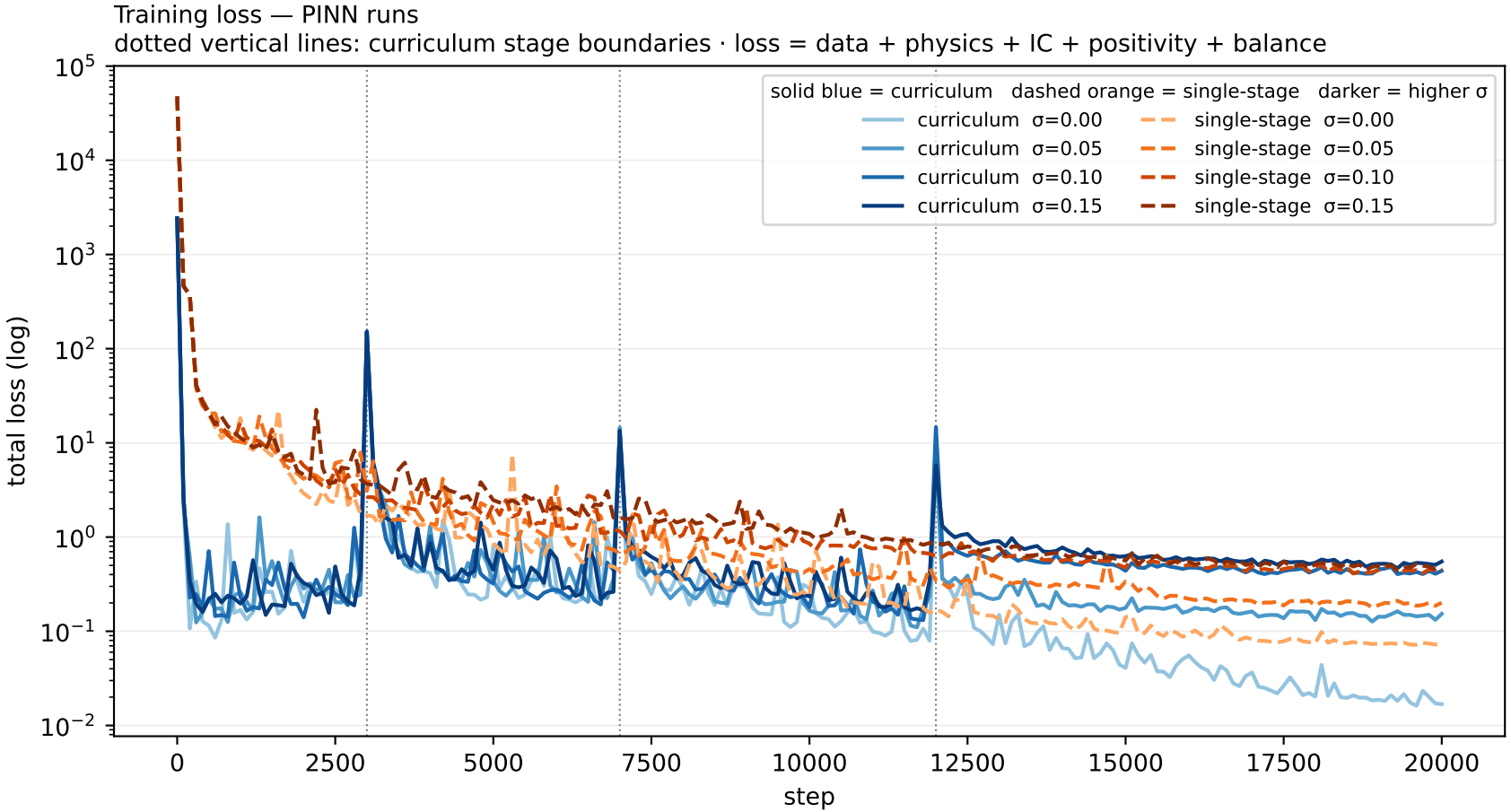
Training loss — PINN runs
dotted vertical lines: curriculum stage boundaries · loss = data + physics + IC + positivity + balance
solid blue = curriculum dashed orange = single-stage darker = higher σ
curriculum σ=0.00
curriculum σ=0.05
curriculum σ=0.10
curriculum σ=0.15
single-stage σ=0.00
single-stage σ=0.05
single-stage σ=0.10
single-stage σ=0.15
total loss (log)
step
10^{5}
10^{4}
10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
0
2500
5000
7500
10000
12500
15000
17500
20000